Özge Sürer

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Academic Appointments

Miami University, Oxford, OH
Department of Information Systems and Analytics
Assistant Professor — August 2022–present
G. Richard and Jane S. Thomas Assistant Professor — July 2026–present

Northwestern University, Evanston, IL
Northwestern Argonne Institute of Science and Engineering
Postdoctoral Research Fellow — January 2021–August 2022

Northwestern University, Evanston, IL
Department of Industrial Engineering and Management Sciences
Postdoctoral Scholar — October 2020–December 2020
Graduate Assistant — September 2016–August 2020

Education

- Ph.D., Industrial Engineering and Management Sciences, Northwestern University, 2020.
- M.S., Industrial Engineering and Management Sciences, Northwestern University, 2016.
- M.S., Industrial Engineering, Boğaziçi University, 2014.
- B.S., Industrial Engineering, İstanbul Technical University, 2011.

Research Interests

Design and analysis of physical and computer experiments; active learning; simulation model calibration; validation and uncertainty quantification; surrogate modeling; interpretable statistical learning; digital twins; ensemble learning with multiple simulation models.

Publications

Peer-Reviewed Journal Articles

12. **Özge Sürer**. Active learning for data-efficient calibration of stochastic simulation models. *Journal of Quality Technology*, 2026. (link)

11. **Özge Sürer**. Batch sequential experimental design for calibration of stochastic simulation models. *Technometrics*, 68(1), 1–13, 2026. (link)

10. **Özge Sürer**. Simulation experiment design for calibration via active learning. *Journal of Quality Technology*, 57(1), 16–34, 2025. (link) [*Invited to JQT Invited Session at QPRC 2025*]

9. Sara Shashaani, **Özge Sürer**, Matthew Plumlee, Seth Guikema. Building trees for probabilistic prediction via scoring rules. *Technometrics*, 66(4), 625–637, 2024. (link) [*Invited to Technometrics Invited Session at FTC 2024*]

8. **Özge Sürer**, Matthew Plumlee, Stefan M. Wild. Sequential Bayesian experimental design for calibration of expensive simulation models. *Technometrics*, 66(2), 157–171, 2024. (Published Online: 2023) (link)

7. **Özge Sürer**, Daniel W. Apley, Edward C. Malthouse. Discovering interpretable structure in longitudinal data via coefficient trees. *Advances in Data Analysis and Classification*, 18, 911–951, 2024. (Published Online: 2023) (link)

6. **Özge Sürer**, Daniel W. Apley, Edward C. Malthouse. Coefficient tree regression: fast, accurate and interpretable predictive modeling. *Machine Learning*, 113(7), 4723–4759, 2024. (Published Online: 2021) (link)

5. Dan Liyanage, **Özge Sürer**, Matthew Plumlee, Stefan M. Wild, Ulrich Heinz. Bayesian calibration of viscous anisotropic hydrodynamic simulations of heavy ion collisions. *Physical Review C*, 108, 054905, 2023. (link)

4. **Özge Sürer**, Filomena M. Nunes, Matthew Plumlee, Stefan M. Wild. Uncertainty quantification in breakup reactions. *Physical Review C*, 106, 024607, 2022. (link)

3. **Özge Sürer**, Daniel W. Apley, Edward C. Malthouse. Coefficient tree regression for generalized linear models. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, 14, 407–429, 2021. (link)

2. Haoxiang Yang, **Özge Sürer**, Daniel Duque, David P. Morton, Bismark Singh, Spencer Fox, Remy Pasco, Kelly Pierce, Paul Rathouz, Zhanwei Du, Michael Pignone, Mark E. Escott, Stephen I. Adler, S. Clairborne Johnston, Lauren Ancel Meyers. Design of COVID-19 staged alert systems to ensure healthcare capacity with minimal closures. *Nature Communications*, 12, 3767, 2021. (link)

1. Seda Yanik, **Özge Sürer**, Başar Öztayşi. Designing sustainable energy regions using genetic algorithms and location-allocation approach. *Energy*, 161–172, 2016. (link)

Working Papers

3. **Özge Sürer**, Xi Chen, Sara Shashaani. Adaptive digital twin calibration via weighted conformal prediction and evidence processes. [Major revision under review at *INFORMS Journal on Data Science*].

2. **Özge Sürer**, Matthew T. Pratola. Experimental design for Bayesian tree-based mixing of simulation models. [Work in progress].

1. Daniel R. Phillips, . . . , **Özge Sürer**, . . . , John C. Yannotty. Solving nuclear physics problems using the Bayesian analysis of nuclear dynamics (BAND) framework. [Work in progress].

Peer-Reviewed Conference Proceedings

6. Xi Chen, **Özge Sürer**. Stochastic simulation metamodeling: Gaussian process methods, adaptive design, and emerging directions. Accepted at *Proceedings of the Winter Simulation Conference*, 2026.

5. **Özge Sürer**, Xi Chen, Sara Shashaani. A dynamic calibration framework for digital twins using active learning and conformal prediction. *Proceedings of the Winter Simulation Conference*, 2025. (link)

4. **Özge Sürer**, Stefan M. Wild. An active learning performance model for parallel Bayesian calibration of expensive simulations. *NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty*, 2024. (link)

3. **Özge Sürer**, Matthew Plumlee. Calibration using emulation of filtered simulation results. *Proceedings of the Winter Simulation Conference*, 2021. (link)

2. **Özge Sürer**, Robin Burke, Edward C. Malthouse. Multistakeholder recommendation with provider constraints. *Proceedings of the 12th ACM Conference on Recommender Systems*, 54–62, 2018. (link)

1. **Özge Sürer**. Improving similarity measures using ontological data. *Proceedings of the 11th ACM Conference on Recommender Systems*, 416–420, 2017. (link)

Book Chapters

2. **Özge Sürer**, Sezi Çevik Onar, İlker Topçu. Innovation strategy evaluation process using fuzzy cognitive mapping. *Intelligent Techniques in Engineering Management*, 107–128, 2015. (link)

1. Başar Öztayşi, **Özge Sürer**. Supply chain performance measurement using a SCOR based fuzzy VIKOR approach. *Supply Chain Management Under Fuzziness*, 199–224, 2014. (link)

Technical Reports

6. Lois Curfman McInnes, . . . , **Özge Sürer**, . . . , Lingfei Wu. Report of the 2025 Workshop on Next-Generation Ecosystems for Scientific Computing: Harnessing Community, Software, and AI for Cross-Disciplinary Team Science, 2025. (link)

5. Matthew Plumlee, **Özge Sürer**, Stefan M. Wild. Surmise users manual, Version 0.1.0, NAISE, 2021. (link)

4. Paul J. Rathouz, Victoria Valencia, Patrick Chang, David P. Morton, Haoxiang Yang, **Özge Sürer**, Spencer J. Fox, Elizabeth C. Matsui, Alex B. Haynes, Lauren A. Meyers. Survival analysis methods for analysis of hospitalization data: Application to COVID-19 patient hospitalization experience. Technical Report, 2021. (link)

3. Nazlican Arslan, **Özge Sürer**, David P. Morton, Haoxiang Yang, Michael Lachmann, Spencer Woody, Spencer J. Fox, Lauren Ancel Meyers. COVID-19 alert stages, healthcare projections and mortality patterns in Austin, Texas. Technical Report, 2021. (link)

2. Haoxiang Yang, Michael Lachmann, **Özge Sürer**, Spencer J. Fox, David P. Morton, Lauren Ancel Meyers. Projecting need for a COVID-19 alternate care site (ACS) in Austin. Technical Report, 2021. (link)

1. Haoxiang Yang, Daniel Duque, **Özge Sürer**, David P. Morton, Remy Pasco, Kelly Pierce, Spencer Fox, Lauren Ancel Meyers. Staged strategy to avoid hospital surge and preventable mortality, while reducing the economic burden of social distancing measures. Technical Report, 2020. (link)

Research Funding

Awarded External Grants

1. **National Science Foundation**, *Advancing Theory for Nuclear Double-Beta Decay (@NBD)*. Duration: 2024-2029. Role: Senior Personnel. Total project award: $2,000,000 (Miami University share $57,800).
2. **National Science Foundation**, *Bayesian Analysis of Nuclear Dynamics (BAND)*. Duration: 2022-2026. Role: Senior Personnel. Total project award: $3,716,619 (Miami University share $169,021).

Awarded Internal Grants

1. **FSB Summer Research Grant**, *A Framework for Calibrating Digital Twins*. Duration: 2026. Award: $10,000.
2. **Miami University Advanced Research Teams (ART) Grant program**, *Examining the Privacy Implications from the Use of Wearables*. Duration: 2023-2024. Role: PI (with three co-PIs). Total project award: $59,339.

Pending Proposal

1. **National Science Foundation**, *A Cyberinfrastructure Framework for Experimental Design in Nuclear Physics (BAND-X)*. Duration: 2026-2031. Role: Miami's institutional PI. Total requested: $5,000,000 (Miami University requested share $385,000).

Awards

- **James Robeson Junior Faculty Award for Research Excellence**, Farmer School of Business, Miami University, 2026.
- **Travel Award**, Conference: Information and Statistics in Nuclear Experiment and Theory, Trento, Italy, 2025.
- **Travel Award**, Workshop: Toward Next-Generation Ecosystems for Scientific Computing, Chicago, IL, 2025.
- **Travel Award**, Conference: Center for Approximation and Mathematical Data Analytics, College Station, TX, 2023.

Open-Source Software

1. `BAND Framework`, v1.0.0 released October 2026, [coauthor] website, project page
   Set of publicly available software tools—a Cyberinfrastructure Framework—designed to facilitate principled uncertainty quantification (UQ) with multiple physics models.
2. `PUQ`, v0.1.1 released October 2025, [maintainer/coauthor] read the docs, project page
   Python package for parallel implementation of novel uncertainty quantification methods.
3. `surmise`, v1.0.0 released September 2026, [coauthor] read the docs, project page
   Python package designed to provide a surrogate model interface for calibration, uncertainty quantification, and sensitivity analysis.
4. `CTR`, released September 2020, [maintainer/coauthor] vignette, project page
   R package for the application of coefficient tree regression (CTR).

Teaching Experience

All courses carry three credit hours. Section enrollments are listed in parentheses.

Miami University

- **Statistical Monitoring and Design of Experiments (ISA/STA 365)**: Fall 2026 (30, 34), Spring 2026 (36, 35). Two sections each semester.
- **Applied Regression Analysis (ISA 391)**: Fall 2025, two sections (27, 28).

- **Applied Regression Analysis (ISA 291)**: Fall 2024 (25, 32); Spring 2024 (30, 30); Fall 2023 (26, 26); Spring 2023 (27, 17); Fall 2022 (22, 14). Two sections each semester.

Northwestern University

- **Statistics (IEMS 303)**: Spring 2019, one section (26).

Research Mentoring

Doctoral student

Yuriy Volkotrub June–September 2021
Building a statistical method for quantifying the uncertainty in a physics model
Bayesian Analysis of Nuclear Dynamics Collaboration, Summer Fellow

Undergraduate students

Northwestern University, Industrial Engineering and Management Sciences Department

Justin Chen June–September 2021
Integration of visualization and diagnostics modules to the Python package `surmise`

Huangda Shang (co-supervised with D. Morton) September 2020–March 2021
Alternative solutions of COVID-19 staged alert systems via derivative-free optimization methods

Aneesh Kudaravalli, Katherine Johns, Margot Dupeyroux,

Robert Wong, Yun Hwan Choi (co-supervised with D. Morton) September–December 2020
Investigation of different trigger metrics for COVID-19 staged alert systems

Achyut Kasi, Cindy Sanchez (co-supervised with D. Morton) June–September 2020
Development of a module to produce graphs and reports automatically for the COVID-19 staged alert system

Conference Presentations and Invited Talks

33. (*upcoming*) The Joint Research Conference on Statistics in Quality, Industry, and Technology, Storrs, CT (June, 2027). *Adaptive digital twin calibration via weighted conformal prediction and evidence processes.*

32. (*upcoming*) INFORMS Annual Meeting, San Francisco, CA (November, 2026). *Adaptive digital twin calibration via weighted conformal prediction and evidence processes.*

31. The Annual IMS/ASA Spring Research Conference (SRC), Clemson, SC (May, 2026). *Active learning for data-efficient calibration of stochastic simulation models.*

30. SIAM Conference on Uncertainty Quantification, Minneapolis, MN (March, 2026). *Dynamic calibration framework for digital twins using active learning and conformal prediction.*

29. 2025 Winter Simulation Conference (WSC 2025), Seattle, WA (December, 2025). *A dynamic calibration framework for digital twins using active learning and conformal prediction.*

28. Information and Statistics in Nuclear Experiment and Theory (ISNET-11), at ECT*, Trento, Italy (November, 2025). *Designing computer experiments for calibration.*

27. Fall Meeting of the APS Division of Nuclear Physics, Workshop: Systematic and Model Uncertainties in Nuclear Experiment and Theory: new tools, techniques, and opportunities for collaboration, Chicago, IL (October, 2025). *Designing real and numerical experiments with uncertainties in mind.*

26. North Carolina State University, Industrial and Systems Engineering Seminar Series, Raleigh, NC (September, 2025). *Batch sequential experimental design for calibration of stochastic simulation models.*

25. The Joint Statistical Meetings (JSM), Nashville, TN (August, 2025). *Batch sequential experimental design for calibration of stochastic simulation models.*

24. The Quality and Productivity Research Conference (QPRC), Seattle, WA (June, 2025). *Simulation experiment design for calibration via active learning.*

23. The Annual ASA/IMS Spring Research Conference (SRC), New York, NY (June, 2025). *Batch sequential experimental design for calibration of stochastic simulation models.*

22. INFORMS Annual Meeting, Seattle, WA (October, 2024). *Simulation experiment design for calibration via active learning.*

21. Fall Technical Conference (FTC), Nashville, TN (October, 2024). *Simulation experiment design for calibration via active learning.*

20. The Joint Statistical Meetings (JSM), Portland, OR (August, 2024). *Simulation experiment design for calibration via active learning.*

19. The Design and Analysis of Experiments (DAE) Conference, Blacksburg, VA (May, 2024). *Sequential Bayesian experimental design for calibration of expensive simulation models.*

18. INFORMS Annual Meeting, Phoenix, AZ (October, 2023). *Performance analysis of sequential experimental design for calibration in parallel computing environments.*

17. The Center for Approximation and Mathematical Data Analytics (CAMDA) Conference, College Station, TX (May, 2023). *Sequential Bayesian experimental design for calibration of expensive simulation models.*

16. Information and Statistics in Nuclear Experiment and Theory (ISNET) Conference, St. Louis, MO (May, 2023). *Sequential Bayesian experimental design for calibration of expensive physics models.*

15. 2022 Fall Meeting of the Division of Nuclear Physics of the American Physical Society, New Orleans, LA (October, 2022). *The Bayesian analysis of nuclear dynamics framework.*

14. INFORMS Annual Meeting, Indianapolis, IN (October, 2022). *Batch sequential calibration of a computationally intensive simulation model using parallel computing.*

13. SIAM Conference on Uncertainty Quantification, Atlanta, GA (April, 2022). *A sequential approach to calibration of a computationally intensive model.*

12. SIAM Conference on Parallel Processing for Scientific Computing, Seattle, WA (February, 2022). *Calibration of a computationally intensive model with parallel computing aspects.*

11. Winter Simulation Conference, Phoenix, AZ (December, 2021). *Calibration using emulation of filtered simulation results.*

10. INFORMS Annual Meeting, Anaheim, CA (October, 2021). *Calibration using emulation of filtered simulation results.*

9. INFORMS Workshop on Data Mining and Decision Analytics, Anaheim, CA (October, 2021). *Discovering group structure in longitudinal data.*

8. INFORMS Annual Meeting, Seattle, WA (October, 2019). *Coefficient tree regression for discovering structure in generalized linear models.*

7. INFORMS Annual Meeting, Seattle, WA (October, 2019). *Discovering structure in longitudinal data via coefficient tree regression.*

6. INFORMS Annual Meeting, Phoenix, AZ (November, 2018). *Coefficient tree regression for discovering hidden structure.*

5. The 12th ACM Conference on Recommender Systems, Vancouver, Canada (October, 2018). *Multi-stakeholder recommendation with provider constraints.*

4. The Midwest Machine Learning Symposium, Chicago, IL (June, 2018). *Coefficient tree regression for discovering hidden structure.*

3. The 11th ACM Conference on Recommender Systems, Como, Italy (August, 2017). *Improving similarity measures using ontological data.*

2. The 34th National Conference for Operations Research and Industrial Engineering, Bursa, Turkey (June, 2014). *Event and clock-based representations in mathematical optimization.*

1. The 26th European Conference on Operational Research, Rome, Italy (July, 2013). *Simulated annealing algorithm with variable cluster number and comparison with k-means algorithm.*

Service to the Profession

1. **Conference organizing committee member**, Information and Statistics in Nuclear Experiment and Theory, Lawrence Berkeley National Laboratory, October 5-9, 2026.
2. **Guest editor**, IISE Transactions Special Issue on Data Science for Computational Modeling (DSCM): Design, Uncertainty Quantification, and Optimization, 2025-2026.
3. **Reviewer and Jury member, Student paper competition**, 45th National Congress on Operations Research and Industrial Engineering (OR/IE), Turkey, 2026.
4. **Reviewer, Student paper competition**, American Statistical Association (ASA); Sections on Physical and Engineering Sciences (SPES) and Quality and Productivity (Q&P), 2026.
5. **Advisory board member**, Information and Statistics in Nuclear Experiment and Theory (ISNET), 2023-Present.
6. **Search committee member for the postdoctoral research fellow**, @NBD Collaboration, January 2025.
7. **Mentor for the ASA Q&P Section mentoring program**, October 2024-December 2024.
8. **Session organizer & chair**
   - *Modern Perspectives on Statistical Learning for Uncertainty Quantification*, JSM, Nashville, TN, August, 2025.
   - *Design and Computer Experiments*, Quality and Productivity Research Conference, Seattle, WA. June, 2025.
   - *Computational/Statistical Methods for Uncertainty Quantification*, INFORMS Annual Meeting, Seattle, WA. October, 2024.
   - *Efficient Statistical Methods for Uncertainty Quantification*, INFORMS Annual Meeting, Indianapolis, IN. October, 2022.
   - *Data-driven Modeling in Uncertainty Quantification*, INFORMS Annual Meeting, Anaheim, CA. October, 2021.
   - *Interpretable Predictive Models*, INFORMS Annual Meeting, Seattle, WA. October, 2019.
9. **Minisymposium organizer & chair**

- *Advances in Digital Twin Calibration and Validation*, SIAM Conference on Uncertainty Quantification, Minneapolis, MN. March, 2026.
- *Statistical Methods for Uncertainty Quantification and Parallel Computing*, SIAM Conference on Parallel Processing for Scientific Computing, Seattle, WA. February, 2022.

10. **Session chair**
    - Intriguing Tweaks in Data Science I, INFORMS Annual Meeting, Phoenix, AZ. November, 2018.
11. **Reviewer for journals including:**
    - *ACM Transactions on Modeling and Computer Simulation* ×1
    - *Advances in Data Analysis and Classification* ×2
    - *Annals of Applied Statistics* ×2
    - *Data Science in Science* ×1
    - *IISE Transactions* ×6
    - *Information Systems* ×1
    - *INFORMS Journal on Computing* ×3
    - *INFORMS Journal on Data Science* ×6
    - *International Journal for Uncertainty Quantification* ×2
    - *Journal of Quality Technology* ×5
    - *Machine Learning* ×1
    - *Mathematics* ×1
    - *npj Digital Medicine* ×2
    - *Scientific Reports* ×2
    - *SIAM/ASA Journal on Uncertainty Quantification* ×1
    - *Statistical Science* ×1
    - *Technometrics* ×5
12. **Professional development:**
    - Toward Next-Generation Ecosystems for Scientific Computing, 2025: invited to participate in the workshop
    - INFORMS Junior Faculty Network Session, 2024: selected to present my research as a representative of the Quality, Statistics and Reliability Section
    - INFORMS New Faculty Colloquia, 2023: participated in faculty development sessions and panel discussions.
13. **Community engagement:**
    - Software tutorials and presentations:
      - Presented on `PUQ` at the BAND-Nuclear Data Workshop in December 2024.
      - Delivered the virtual presentation on the `BAND Framework` at the STAR Collaboration's "Spring Juniors Day" in March 2024.
      - Led tutorials (see an example) and interactive presentations (see an example Google Colab doc) on `surmise` during the annual BAND Camps in December 2021 and May 2023.